

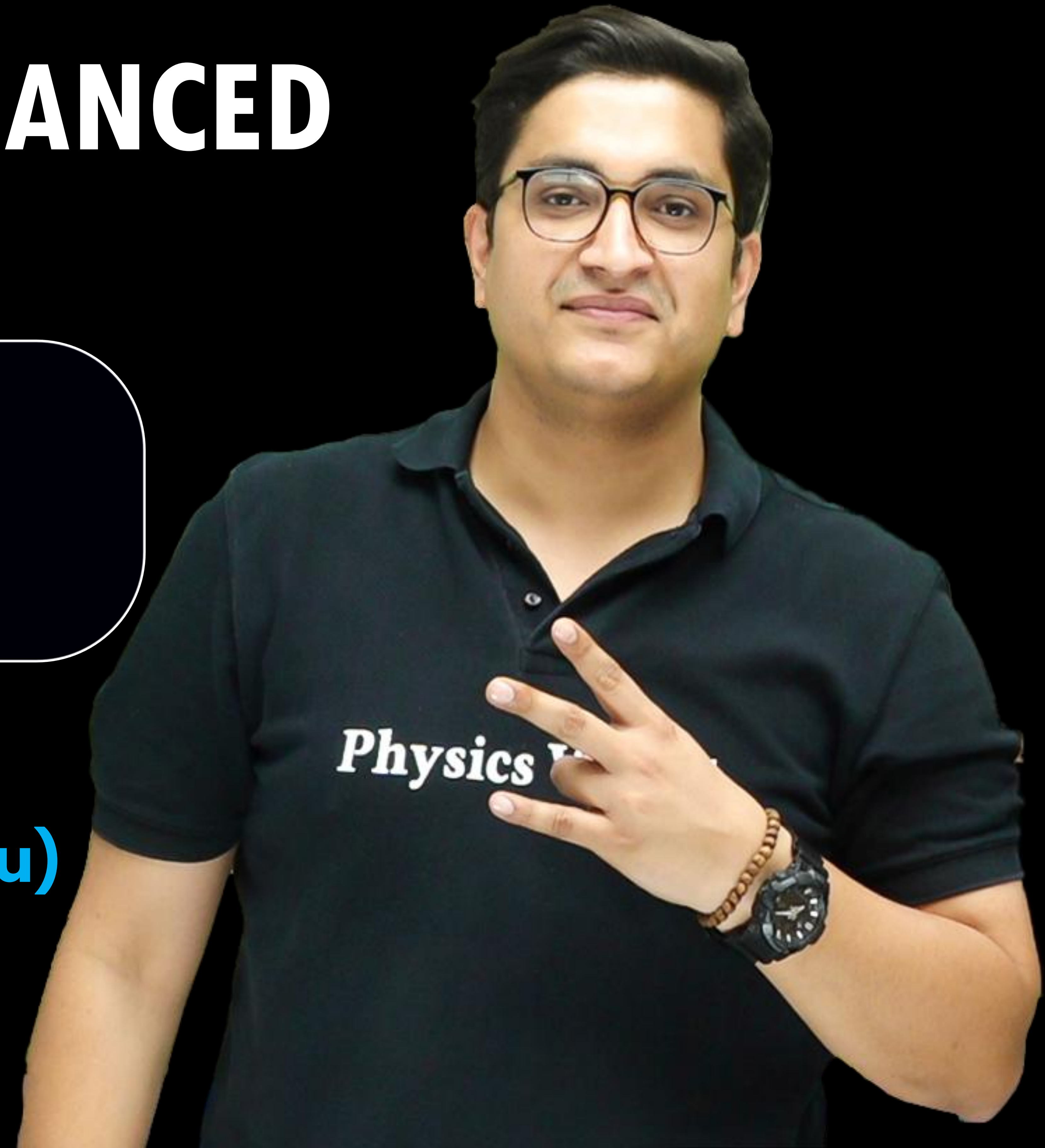
JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

MATRICES

DIBY (DO IT BY YOURSELF)

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DIBY-01

If $A^2 = A$, (A is a square matrix) and $(I + A)^4 = I + kA$ (k is a scalar), then value of k is _____.

[Ans. : 15]

DIBY-02

Matrix A satisfies $A^2 = 2A - I$ where I is the identity matrix and $A^{10} = \alpha A - \beta I$ ($\alpha, \beta \in \text{scalar}$), then value of $\alpha + \beta =$ _____.

[Ans. : 1]

DIBY-03

Let $A = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix}$, then show that $A^n = \begin{bmatrix} a^n & \frac{b(a^n - 1)}{(a - 1)} \\ 0 & 1 \end{bmatrix}$

DIBY-04

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be two matrices such that their product is commutative to each other and $c \neq 3b$, find the value of $\left| \frac{a-d}{3b-c} \right|$.

[Ans. : 01]

DIBY-05

If $A = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then:

[JEE Mains-2023]

(A) $A^{30} - A^{25} = 2I$

(B) $A^{30} + A^{25} + A = I$

(C) $A^{30} + A^{25} - A = I$

(D) $A^{30} = A^{25}$

[Ans. : (C)]

DIBY-06

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & -1 \\ 0 & 12 & -3 \end{bmatrix}$. Then the sum of the diagonal elements of the matrix $(A + I)^{11}$ is equal to:

[JEE Mains-2023]

(A) 6144

(B) 4094

(C) 4097

(D) 2050

[Ans. : (C)]

DIBY-07

If $P = \begin{bmatrix} 1 & 0 \\ \frac{1}{2} & 1 \end{bmatrix}$, then P^{50} is:

[JEE Mains-2021]

(A) $\begin{bmatrix} 1 & 0 \\ 25 & 1 \end{bmatrix}$

(B) $\begin{bmatrix} 1 & 25 \\ 0 & 1 \end{bmatrix}$

(C) $\begin{bmatrix} 1 & 0 \\ 50 & 1 \end{bmatrix}$

(D) $\begin{bmatrix} 1 & 50 \\ 0 & 1 \end{bmatrix}$

[Ans. : (A)]

DIBY-08

If $A = \frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$, then:

[JEE Mains-2023]

(A) $A^{30} - A^{25} = 2I$

(B) $A^{30} = A^{25}$

(C) $A^{30} + A^{25} - A = I$

(D) $A^{30} + A^{25} + A = I$

[Ans. : (C)]

DIBY-09

For the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} -29 & 49 \\ -13 & 18 \end{bmatrix}$, if $(A^{15} + B) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, then among the following which one is true?

(A) $x = 5, y = 7$

(B) $x = 18, y = 11$

(C) $x = 11, y = 2$

(D) $x = 16, y = 3$

[Ans. : (C)]

DIBY-10

Let $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ and B be two matrices such that $A^{100} = 100B + I$. Then the sum of all the elements of B^{100} is _____.

[Ans. : (0)]

DIBY-11

If $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$ and **trace** $A^{13} = 3^m$, then find m .

[Ans. : 7]

DIBY-12 (Challenger)

The set of natural numbers is divided into arrays of rows and columns in the form of

matrices as $A_1 = (1)$, $A_2 = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$, $A_3 = \begin{pmatrix} 6 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{pmatrix}$, so on. Find the value of $T_r(A_{10})$.

[Ans. : 3355]

DIBY-13 (Acidic)

Let A & B are matrices of order 3×3 and $|A| = -2$ & $|B| = 3$, then find value of :

(i) $\det(2AB)$

(ii) $\det(4A^2B^3)$

(iii) $\det(|3AB|A)$

(iv) $|A|2B|B^2|$

[Ans. : (i) -48 , (ii) 6912 , (iii) $2^4 \cdot 3^{12}$, (iv) $-2^{10} \cdot 3^5$]

DIBY-14 (Challenger)

If $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$, then find the value of determinant $(A + A^2B^2 + A^3 + A^4B^4 + \dots \text{upto } 100 \text{ terms})$.

[Ans. : 0]

DIBY-15

A square matrix P satisfies $P^2 = I - P$; where I is an identity matrix. If $P^n = 5I - 8P$, then find the value of n .

[Ans. : 6]

DIBY-16

Let A and B be two square matrices of order 2. If $\det(A) = 2$, $\det(B) = 3$ and $\det((\det(5(\det A)B)A^2)) = 2^a 3^b 5^c$ for some $a, b, c \in N$, then $a + b + c$ is equal to:

[JEE Mains-2022]

(A) 10

(B) 12

(C) 13

(D) 14

[Ans. : (B)]

DIBY-17

The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is **[JEE Mains-2023]**

- (A) 10^9 (B) 9^{10} (C) 10^6 (D) 6^{10}

[Ans. : (C)]

DIBY-18

Let $A = \begin{bmatrix} 1 & a & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$, $a, b \in \mathbb{R}$. If for some $n \in N$, $A^n = \begin{bmatrix} 1 & 48 & 2160 \\ 0 & 1 & 96 \\ 0 & 0 & 1 \end{bmatrix}$, then $n + a + b$ is equal to

[JEE Mains-2022]

[Ans. : 24]

DIBY-19

Let A be a symmetric matrix such that $|A| = 2$ and $\begin{bmatrix} 2 & 1 \\ 3 & \frac{3}{2} \end{bmatrix} A = \begin{bmatrix} 1 & 2 \\ \alpha & \beta \end{bmatrix}$ of trace A is s , then $\frac{\beta s}{\alpha^2}$ is equal to **[JEE Mains-2023]**

[Ans. : 5]

DIBY-20

Let $A = \begin{bmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{bmatrix}$ and matrix B is defined such that $B = A + 4A^2 + 6A^3 + 4A^4 + A^5$. If $\det(B) = 1$, then find the number of values of α in $[-2\pi, 2\pi]$.

[Ans. : 4]

DIBY-21

Let $A = [a_{ij}]_{3 \times 3}$, $B = [b_{ij}]_{3 \times 3}$ and $C = [c_{ij}]_{3 \times 3}$ be any three matrices, where $b_{ij} = 3^{i-j} \cdot a_{ij}$ and $c_{ij} = 4^{i-j} \cdot b_{ij}$. If $\det A = 2$, then $\det(BC)$ is equal to

[Ans. : 4]

DIBY-22

The number of symmetric matrices of order 3, with all the entries from the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is

[JEE Mains-2023]

(A) 6^{10}

(B) 10^6

(C) 9^{10}

(D) 10^9

[Ans. : (B)]



DIBY-23

Let A be a 3×3 real matrix such that $A^2(A - 2I) - 4(A - I) = O$, where I and O are the identity and null matrices, respectively. If $A^5 = \alpha A^2 + \beta A + \gamma I$, where α, β and γ are real constants, then $\alpha + \beta + \gamma$ is equal to: **[JEE Mains-2025]**

- (A) 12 (B) 20 (C) 76 (D) 4

[Ans. : (A)]

DIBY-24

Let the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ satisfy $A^n = A^{n-2} + A^2 - I$ for $n \geq 3$. Then the sum of all the elements of A^{50} is **[JEE Mains-2025]**

- (A) 53 (B) 52 (C) 39 (D) 44

[Ans. : (A)]

DIBY-25

The number of singular matrices of order 2, whose elements are from the set $\{2, 3, 6, 9\}$ is **[JEE Mains-2025]**

_____.

[Ans. : 36]

DIBY-26

If $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & -2 \\ 0 & 1 \end{bmatrix}$, $P = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$, $B = PAP^T$ and $C = PTB^{10}P$,

then sum of diagonal elements of C is $\frac{m}{n}$, then $m + n$ is equal to

- (A) 127 (B) 258 (C) 65 (D) 2049

[JEE Mains-2025]

[Ans. : (C)]

DIBY-27

Let α be a solution of $x^2 + x + 1 = 0$, and for some a and b in $\mathbb{R}$, $\begin{bmatrix} 4 & a & b \end{bmatrix} \begin{bmatrix} 1 & 16 & 13 \\ -1 & -1 & 2 \\ -2 & -14 & -8 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$. If $\frac{4}{\alpha^4} + \frac{m}{\alpha^a} + \frac{n}{\alpha^b} = 3$, then $m + n$ is equal to

- (A) 3 (B) 11 (C) 7 (D) 8

[JEE Mains-2025]

[Ans. : (B)]

DIBY-28

If $A = \begin{bmatrix} \sqrt{2} & 1 \\ -1 & \sqrt{2} \end{bmatrix}$, $B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $C = ABA^T$ and $X = ATC^2A$, then $\det X$ is equal to

[JEE Mains-2024]

(A) 243

(B) 729

(C) 27

(D) 891

[Ans.: (B)]

DIBY-29

Let $A = \begin{bmatrix} a & x & p \\ y & q & b \\ r & c & z \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ where $a, b, c, x, y, z, p, q, r$ are natural numbers. If $\text{tr.}(AB + AB^3 + AB^5 + \dots + AB^{19}) = 210$, then find number of ordered triplets (p, q, r) .

[Ans.: 190]

DIBY-30

Find adjoint matrices of following matrices

(i) $P = \begin{bmatrix} -2 & 4 \\ 1 & 7 \end{bmatrix}$ (ii) $Q = \begin{bmatrix} -1 & 2 & -3 \\ 4 & 0 & 3 \\ 1 & 2 & -1 \end{bmatrix}$ (iii) $R = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 2 \\ -2 & 3 & 4 \end{bmatrix}$

[Ans.: (i) $\begin{bmatrix} 7 & -4 \\ -1 & -2 \end{bmatrix}$ (ii) $\begin{bmatrix} -6 & -4 & 6 \\ 7 & 4 & -9 \\ 8 & 4 & -8 \end{bmatrix}$ (iii) $\begin{bmatrix} -10 & -8 & 4 \\ -8 & -4 & +2 \\ 1 & -1 & -1 \end{bmatrix}$]

DIBY-31

Let A be a 3×3 real matrix. If $\det(2\text{adj}(2\text{adj}(\text{adj}(2A)))) = 2^{41}$, then the value of $\det(A^2)$ equals

[JEE Mains-2021]

[Ans.: 4]

DIBY-32

Let A and B be two 3×3 matrices such that $AB = I$ and $|A| = \frac{1}{8}$, then $|\text{adj}(B\text{adj}(2A))|$ is equal to

[JEE Mains-2022]

(A) 16

(B) 32

(C) 64

(D) 128

[Ans.: (C)]

DIBY-33

If A is a 3×3 matrix and $|A| = 2$, then $|3\text{adj}(|3A|A^2)|$ is equal to

[JEE Mains-2023]

- (A) $3^{11} \cdot 6^{10}$ (B) $3^{12} \cdot 6^{10}$ (C) $3^{10} \cdot 6^{11}$ (D) $3^{12} \cdot 6^{11}$

[Ans. : (A)]

DIBY-34

Let $A = \begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$. If $B = I - {}^5C_1(\text{adj } A) + {}^5C_2(\text{adj } A)^2 - \dots - {}^5C_5(\text{adj } A)^5$, then the sum of all elements of the matrix B is

[JEE Mains-2022]

- (A) -5 (B) -6 (C) -7 (D) -8

[Ans. : (C)]

DIBY-35

Let for $A = \begin{bmatrix} 1 & 2 & 3 \\ a & 3 & 1 \\ 1 & 1 & 2 \end{bmatrix}$, $|A| = 2$. If $|2\text{adj}(2\text{adj}(2A))| = 32^n$, then $3n + \alpha$ is equal to

[JEE Mains-2023]

- (A) 10 (B) 9 (C) 12 (D) 11

[Ans. : (D)]

DIBY-36

If $A = \frac{1}{5!6!7!} \begin{bmatrix} 5! & 6! & 7! \\ 6! & 7! & 8! \\ 7! & 8! & 9! \end{bmatrix}$, then $|\text{adj}(\text{adj}(2A))|$ is equal to:

[JEE Mains-2023]

- (A) 2^8 (B) 2^{12} (C) 2^{20} (D) 2^{16}

[Ans. : (D)]

DIBY-37

The number of all 3×3 matrices A , with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3 is _____

[JEE Mains-2020]

[Ans. : 672]

DIBY-38

Let M be any 3×3 matrix with entire from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven is

[JEE Mains-2021]

[Ans. : 540]

DIBY-39

If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj}(3A^2 + 12A)$ is equal to:

[JEE Mains-2017]

(A) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

(B) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$

(C) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

(D) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$

[Ans. : (C)]

DIBY-40

$A = \begin{bmatrix} -1 & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 4 & -1 \end{bmatrix}$ compute A^{-1} and $|A^{-1}|$.

[Ans. : $\frac{-1}{2} \begin{bmatrix} 7 & 4 & -1 \\ -4 & -2 & 0 \\ 5 & 4 & -1 \end{bmatrix}, \frac{-1}{2}$]

DIBY-41

Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix}$.

If $AB = I$, where I is an identity matrix of order 3, then find $\text{tr}(B)$.

[Ans. : $\frac{2}{5}$]

DIBY-42

Let A be matrix of order 3×3 such that $|A| = 1$. Let $B = 2A^{-1}$ and $C = \frac{\text{adj}A}{\sqrt[3]{2}}$, then find the value of $|AB^2C^3|$. (where $|A|$ represent $\det. A$)

[Ans. : 8]

DIBY-43

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to:

[JEE Mains-2019]

(A) 10

(B) 15

(C) 135

(D) 9

[Ans. (A)]

DIBY-44

Let $A = \{a_{ij}\}$ be a 3×3 matrix, where $a_{ij} = \begin{cases} (-1)^{j-i} & \text{if } i < j \\ 2 & \text{if } i = j \\ (-1)^{i+j} & \text{if } i > j \end{cases}$, then $\det(3\text{Adj}(2A^{-1}))$ is equal to.....

[JEE Mains-2021]

[Ans. : 108]

DIBY-45

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to

[JEE Mains-2021]

[Ans. : 16]

DIBY-46

Let A be a 3×3 matrix such that $|\text{adj}(\text{adj}(\text{adj}A))| = 81$.

If $S = \left\{ n \in \mathbb{Z} : (|\text{adj}(\text{adj}A)|)^{\frac{(n-1)^2}{2}} = |A|^{(3n^2-5n-4)} \right\}$, then $\sum_{n \in S} |A^{(n^2+n)}|$ is equal to

[JEE Mains-2025]

(A) 866

(B) 750

(C) 820

(D) 732

[Ans. : (D)]

DIBY-47

Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1}B^T)$ is equal to: [JEE Main-2019]

- (A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{16}$ (D) 16

[Ans. : (c)]

DIBY-48

Show that $A^{-1} = \frac{1}{4}(-A^2 + 2A + 3I)$ where matrix $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$

DIBY-49

If P is orthogonal matrix and A periodic matrix with period with 4, $Q = PAP^T$, then find $P^T Q^{2005} P$.

[Ans. :]



DIBY-50

If A is a non singular (2×2) matrix such that $A + \text{adj } A = A^{-1}$, then find the value of $|2A^{-1}|$.
[Ans. : 8]

DIBY-51

Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$.

Let $A^{-1} = xA^2 + yA + zI$ then find the value of $(x + y + z)$ where I is a unit matrix of order 3.
[Ans. : 01]

DIBY-52

$A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal find $2\alpha^2 + 6\beta^2 + 3\gamma^2$.

[Ans. : 03]

DIBY-53

Let A be a 3×3 matrix such that $\text{adj } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and $B = \text{adj}(\text{adj} A)$.

If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair, $(|\lambda|, \mu)$ is equal to [JEE Mains-2020]

- (A) $(3, 81)$ (B) $(9, \frac{1}{9})$ (C) $(3, \frac{1}{81})$ (D) $(9, \frac{37}{16})$

[Ans.: (C)]

DIBY-54

Let α be a root of the equation $x^2 + x + 1 = 0$ and the matrix $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha^4 \end{bmatrix}$, then the

matrix A^{31} is equal to

- (A) A^3 (B) I_3 (C) A (D) A^2

[JEE Mains-2020]

[Ans.: (A)]



DIBY-55

Consider the matrices $A = \begin{bmatrix} 2 & -2 \\ 4 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 9 \\ 1 & 3 \end{bmatrix}$. If matrices P and Q are such that $PA = B$ and $AQ = B$, then the absolute value of the sum of the diagonal elements of $2(P + Q)$ is ____.

[JEE Mains-2026]

[Ans. : 34]

DIBY-56

Let $A = \begin{bmatrix} 1 & 1 & 2 \\ -2 & 0 & 1 \\ 1 & 3 & 5 \end{bmatrix}$, then the sum of all elements of the matrix $\text{adj}(\text{adj}(2(\text{adj}A)^{-1}))$ is equal to

[JEE Mains-2026]

- (A) 3 (B) 4 (C) -4 (D) -3

[Ans. : (D)]

DIBY-57

Let $P = [p_{ij}]$ and $Q = [q_{ij}]$ be two square matrices of order 3 such that $q_{ij} = 2^{(i+j-1)}p_{ij}$ and $\det(Q) = 210$. then the value of $\det(\text{adj}(\text{adj}(P)))$ is

[JEE mains-2026]

- (A) 81 (B) 16 (C) 124 (D) 32

[Ans. : (B)]

DIBY-68

Let $S = \left\{ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} : a, b, c, d \in \{0, 1, 2, 3, 4\} \text{ and } A^2 - 4A + 3I = 0 \right\}$ be a set of 2×2 matrices. Then the number of matrices in S , for which the sum of the diagonal elements is equal to 4, is: [JEE Mains-2026]

- (A) 20 (B) 17 (C) 21 (D) 19

[Ans.: (D)]

DIBY-69

Let $A = \begin{bmatrix} 1 & 2 \\ 1 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ \beta & 2 \end{bmatrix}$. If $A^2 - 4A + I = 0$ and $B^2 - 5B - 6I = 0$, then among the two statements:

$S_1 : [(B - A)(B + A)]^T = \begin{bmatrix} 13 & 15 \\ 7 & 10 \end{bmatrix}$ and $S_2 : \det(\text{adj}(A + B)) = -5$, [JEE Mains-2026]

- (A) only S_1 is correct (B) only S_2 is correct
(C) both S_1 and S_2 are correct (D) both S_1 and S_2 are wrong

[Ans.: (B)]



DIBY-70

For some $\alpha, \beta \in \mathbb{R}$, let $A = \begin{bmatrix} \alpha & 2 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & \beta \end{bmatrix}$ be such that $A^2 - 4A + 2I = B^2 - 3B + I = 0$.

Then $\left(\det\left(\text{adj}(A^3 - B^3)\right)\right)^2$ is equal to

[JEE Mains-2026]

[Ans. : (225)]

DIBY-71

Let A be a non-singular square matrix of order 2, such that $|A + |A| \text{adj}(A)| = 0$, then the value of $|A - |A| \text{adj}(A)| = \underline{\hspace{2cm}}$.

[Ans. : (4)]

DIBY-72

Let A and B be two square matrices of order 3 such that $|A| = 3$ and $|B| = 2$.

Then $|A^T A (\text{adj}(2A))^{-1} (\text{adj}(4B)) (\text{adj}(AB))^{-1} A A^T|$ is equal to:

[JEE Mains-2024]

(A) 22

(B) 24

(C) 26

(D) 20

[Ans. : (64)]

DIBY-73

If $A = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, then the determinant of the matrix $(A^{2025} - 3A^{2024} + A^{2023})$ is:

[JEE Mains-2026]

[Ans. : 16]

DIBY-74

Let A be a 2×2 real matrix and I be the identity matrix of order 2. If the roots of the equation $|A - xI| = 0$ be -1 and 3 , then the sum of the diagonal elements of the matrix A^2 is

[JEE Mains-2024]

[Ans. : 10]

DIBY-75

Let P be a 2×2 matrix such that $P \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$ and $P^2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$. If x_1 and x_2 are two values of x for which $|P - xI| = 0$, where I is an identity matrix of order 2, then find the value of $x_1^2 + x_2^2$.

[Ans. : 5]

DIBY-76

Let β be a real number. Consider the matrix $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$. If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is

[JEE (Adv.)-2022]

[Ans. : 3]

DIBY-77

Let A be a 3×3 matrix with $\det(A) = 4$. Let R_i denote the i^{th} row of A . If a matrix B is obtained by performing the operation $R_2 \rightarrow 2R_2 + 5R_3$ on $2A$, then $\det(B)$ is equal to

[JEE Mains-2021]

(A) 64

(B) 16

(C) 128

(D) 80

[Ans. : (A)]

DIBY-78

For a 3×3 matrix M , let $\text{trace}(M)$ denote the sum of all the diagonal elements of M . Let A be a 3×3 matrix such that $|A| = \frac{1}{2}$ and $\text{trace}(A) = 3$. If $B = \text{adj}(\text{adj}(2A))$, then the value of $|B| + \text{trace}(B)$ equals **[JEE Mains-2025]**

- (A) 56 (B) 280 (C) 132 (D) 174

[Ans. : (B)]

DIBY-79

Let I be the identity matrix of order 3×3 and for the matrix $A = \begin{bmatrix} \lambda & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -1 & 2 \end{bmatrix}$, $|A| = -1$. Let B be the inverse of the matrix $\text{adj}(A \text{adj}(A^2))$. Then $|(\lambda B + I)|$ is equal to **[JEE Mains-2025]**

[Ans. : 38]

DIBY-80

Let A be a 3×3 real matrix such that $A \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} = 4 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$, $A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Then,

the system $(A - 3I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ has

[JEE Mains-2024]

(A) unique solution

(B) exactly two solutions

(C) no solution

(D) infinitely many solutions

[Ans. : (A)]

DIBY-81

If a point satisfying $(\alpha \ \beta \ \gamma) \begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0 \ 0 \ 0)$ lies on the plane $2x + 4y + 3z = 5$,

then $6\alpha + 9\beta + 7\gamma$ is equal to

[JEE Mains-2023]

A) -1

(B) $\frac{11}{5}$

(C) $\frac{5}{4}$

(D) 11

[Ans. : (D)]

DIBY-82

Determine the product $\begin{bmatrix} -4 & 4 & 4 \\ -7 & 1 & 3 \\ 5 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 1 \\ 1 & -2 & -2 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to solve the system of equations $x - y + z = 4, x - 2y - 2z = 9, 2x + y + 3z = 1$. **[Boards-2017, 6 M]**

[Ans. : $x = 3, y = -2, z = -1$]

DIBY-83

If $A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}$, find A^{-1} and use it to solve the following system of equation:

$5x - y + 4z = 5, 2x + 3y + 5z = 2, 5x - 2y + 6z = -1$ **[Boards-2020, 6M]**

[Ans. $x = 3, y = 2, z = -2$]

DIBY-84

Paragraph

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$ and U_1, U_2 and U_3 are columns of a 3×3 matrix U . If column matrices

U_1, U_2 and U_3 satisfying $AU_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, AU_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, AU_3 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$ evaluate as directed in the

following equations.

[IITJEE-2006]

Q. The value of $|U|$ is

[Ans. : 3]

Q. The sum of the elements of the matrix U^{-1} is

[Ans. : 0]

Q. The determinant value of $\begin{bmatrix} 3 & 2 & 0 \end{bmatrix} U \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$ is

[Ans. : 5]

DIBY-85

Let $S = \left\{ \begin{pmatrix} -1 & a \\ 0 & b \end{pmatrix} : a, b \in \{1, 2, 3, \dots, 100\} \right\}$ and let $T_n = \{A \in S : A^{n(n+1)} = I\}$. Then the number of elements in $\bigcup_{n=1}^{100} T_n$ is ____.

[JEE Mains-2022]

[Ans. : 100]

DIBY-86

There exist a matrix Q such that $PQP^T = N$ where $P = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Given N is a diagonal matrix of the form $N = \text{diag}\{n_1, n_2, n_3\}$ where n_1, n_2 and n_3 are roots of the equation $\det(P - nI) = 0$, ($n_1 < n_2 < n_3$), then find the following:

- (1) $\det(\text{adj}N)$ (2) find the matrix Q (3) $\text{Tr}(P^{2026})$

$$[\text{Ans. : 9, } Q = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 \\ 0 & 0 & 3 \end{bmatrix}, 3^{2026} + 2]$$

DIBY-87

Let $S = \left\{ n \in N : \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}^n \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \forall a, b, c, d \in R \right\}$, where $i = \sqrt{-1}$. Then the number of 2-digit numbers in the set S is

[JEE Mains-2021]

[Ans. : 11]

DIBY-88

The total number of matrices $A = \begin{pmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{pmatrix}$, $(x, y \in R, x \neq y)$ for which $A^T A = 3I_3$ is

[JEE Mains-2019]

(A) 6

(B) 2

(C) 3

(D) 4

[Ans. : (D)]

DIBY-89

Let A be a 3×3 matrix of non-negative real elements such that $A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Then the maximum value of $\det(A)$ is _____.

[JEE Mains-2024]

[Ans. : 27]

DIBY-90

Let A be a non-singular matrix of order 3. If $\det \left(3 \operatorname{adj} \left(2 \operatorname{adj} ((\det A)A) \right) \right) = 3^{-13} \cdot 2^{-10}$ and $\det(3 \operatorname{adj}(2A)) = 2^m \cdot 3^n$, then $|3m + 2n|$ is equal to ____.

[JEE Mains-2024]

[Ans. : 14]

DIBY-91

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$ be a given matrix. If $P = A^4 + 8A$ and $Q = -5A^3 + 8A^2 - 10A$, then

(A) $\operatorname{tr.}(P + Q) = 10$ (B) $\operatorname{tr.}(P + Q) = 0$ (C) $\det.(P + Q) = 8$ (D) $\det.(P + Q) = 32$

[Ans. : A, D]

DIBY-92

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & \alpha \end{bmatrix}$ and $B = \begin{bmatrix} \beta & 1 \\ 1 & 0 \end{bmatrix}$, $\alpha, \beta \in \mathbb{R}$. Let α_1 be the value of α which satisfies

$(A + B)^2 = A^2 + \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$ and α_2 be the value of α which satisfies $(A + B)^2 = B^2$. Then $|\alpha_1 - \alpha_2|$ is equal to

[JEE Mains-2022]

[Ans. : 2]

DIBY-93

If $A = \begin{bmatrix} e^t & e^{-t}\cos t & e^{-t}\sin t \\ e^t & -e^{-t}\cos t - e^{-t}\sin t & -e^{-t}\sin t + e^{-t}\cos t \\ e^t & 2e^{-t}\sin t & -2e^{-t}\cos t \end{bmatrix}$. Then A is [JEE Mains-2019]

- (A) Invertible only if $t = \frac{\pi}{2}$ (B) Not invertible for any $t \in R$
 (C) Invertible for all $t \in R$ (D) Invertible only if $t \in \pi$

[Ans. : (C)]

DIBY-94

The number of matrices of order 3×3 , whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____. [JEE Mains-2022]

[Ans. : 282]

DIBY-95

Let $A = [a_{ij}] = \begin{bmatrix} \log_5 128 & \log_4 5 \\ \log_5 8 & \log_4 25 \end{bmatrix}$. If A_{ij} is the cofactor of a_{ij} , $C_{ij} = \sum_{k=1}^2 a_{ik} A_{jk}$, $1 \leq i, j \leq 2$ and $C = [C_{ij}]$, then $8|C|$ is equal to [JEE Mains-2025]

- (A) 242 (B) 288 (C) 262 (D) 222

[Ans. : (A)]

DIBY-96

Let $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ and $B = \begin{bmatrix} bc - a^2 & ca - b^2 & ab - c^2 \\ ca - b^2 & ab - c^2 & bc - a^2 \\ ab - c^2 & bc - a^2 & ac - b^2 \end{bmatrix}$ be two non-singular matrices such that $(A^2 - 2I)B = 0$ where $a > b > c > 0$, then which of the following statement(s) is(are) correct?

(A) $\text{Tr.}(AB) = 6\sqrt{2}$

(B) $\text{Tr.}(AB) = -6\sqrt{2}$

(C) $\det(A - \sqrt{2}B) = 54\sqrt{2}$

(D) $\det(A - \sqrt{2}B) = -54\sqrt{2}$

[Ans. : B, D]

DIBY-97

Find the Eigen values of the matrix $A = \begin{bmatrix} 2026 & 2025 & 2025 \\ 2025 & 2026 & 2025 \\ 2025 & 2025 & 2026 \end{bmatrix}$.

[Ans. : 1, 1, 6076]

DIBY-98

For a real number α , if the system of linear equations $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$

[Ans. : 1]

DIBY-99

Let $M = (a_{ij})$, $i, j \in \{1, 2, 3\}$ be the 3×3 matrix such that $a_{ij} = 1$, If $j + 1$ is divisible by i , otherwise $a_{ij} = 0$. Then which of the following statement(s) is(are) true?

[JEE (Adv.)-2023]

(A) M is invertible

(B) There exists a nonzero column matrix $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and such that $M \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} -a_1 \\ -a_2 \\ -a_3 \end{pmatrix}$

(C) The set $\{x \in \mathbb{R}^3 : MX = 0\} \neq \{0\}$, where $0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

(D) The matrix $(M - 2I)$ is invertible, where I is 3×3 identity matrix

[Ans. : B, C]

DIBY-100

Consider the matrix $P = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$. Let the transpose of a matrix X be denoted by X^T .

Then the number of 3×3 invertible matrices Q with integer entries, such that $Q^{-1} = Q^T$ and $PQ = QP$, is [JEE (Adv.)-2025]

- (A) 32 (B) 8 (C) 16 (D) 24

[Ans. : (C)]

DIBY-101

Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then:

[JEE Mains-2022]

- (A) The system of linear equations $AX = 0$ has a unique solution.
 (B) The system of linear equations $AX = 0$ has infinitely many solutions.
 (C) B is an invertible matrix.
 (D) $\text{adj}(A)$ is an invertible matrix.

[Ans. : (B)]

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians